

1. Overcoming the Technician Shortage and Reducing Aircraft Downtime

- **Issue:** PAC faces an industry-wide shortage of certified aviation technicians and unprecedented Maintenance, Repair, and Overhaul (MRO) demand to service over 3,000 aircraft, increasing the risk of costly "Aircraft on Ground" (AOG) events 1-3.
- **Action:** We can implement Red Hat Event-Driven Ansible (EDA) integrated with ServiceNow and NetBox via webhooks to automate IT service management, F5 load balancer provisioning, and MRO configuration changes asynchronously the moment approvals are granted 3-5.
- **Value:** This drastically reduces aircraft downtime (AOG) and increases engineering velocity by replacing manual polling with proactive automation, directly mitigating the technician shortage and ensuring 100% on-time delivery from the Customer Delivery Group 3, 4, 6, 7.

2. Escaping Legacy Virtualization Costs to Fund Cloud Innovation

- **Issue:** To achieve its strategic pivot toward becoming a cloud-native SaaS/PaaS provider, PAC must migrate legacy monolithic applications to the cloud and escape ballooning legacy virtualization (VMware) licensing costs post-Broadcom acquisition 8-10.
- **Action:** We provide Red Hat OpenShift Virtualization as a unified hybrid-cloud platform that enables PAC to run existing virtual machines alongside modern containers on the exact same infrastructure 9-11.
- **Value:** This delivers a rapid reduction in infrastructure costs by eliminating redundant legacy hypervisor fees, seamlessly migrating workloads to Azure or AWS, and freeing up capital to improve Return on Equity (ROE) and fund high-margin digital innovation 6, 9, 10, 12.

3. Enabling Flawless Astrova Edge Deployments for Passenger Engagement

- **Issue:** PAC is aggressively expanding its modular Astrova in-flight engagement platform to increase Passenger Engagement Minutes (PEMs), but deploying localized AI and software updates in disconnected aircraft environments carries high risks for patching errors 6, 8, 13, 14.
- **Action:** We deliver RHEL 10 Image Mode and Red Hat Device Edge, allowing PAC to treat the entire operating system as an immutable, tamper-proof container built through CI/CD pipelines 13-15.
- **Value:** This virtually eliminates deployment downtime by enabling instant, smartphone-like rollbacks if an update fails in the field, drastically reducing the time required for FAA/EASA compliance sign-offs while ensuring uninterrupted passenger engagement to drive airline ancillary revenue 6, 13-15.

4. Securing LEO Connectivity and Achieving EASA Part-IS Compliance

- **Issue:** PAC must secure massive volumes of passenger data flowing across new Low Earth Orbit (LEO) satellite networks and meet the strict new EASA Part-IS cybersecurity mandate by 2026 to prevent potential aircraft groundings 13, 16, 17. Additionally, aging legacy Java apps present vulnerabilities 11, 18.
- **Action:** We offer Red Hat OpenShift and Universal Base Image (UBI) to isolate vulnerable legacy applications, combined with RHEL 10 featuring out-of-the-box, FIPS-compliant quantum cryptography 11, 16, 18, 19.

- **Value:** This ensures long-term regulatory compliance and a zero-breach data posture, dramatically improving internal cybersecurity scores, protecting passenger data, and retaining crucial security waivers for uninterrupted airline operations 11, 16-18.

5. Standardizing Engineering Culture and Eliminating Shadow IT

- **Issue:** PAC is working to overcome technical debt caused by a historically fragmented "shadow IT" engineering culture, where developers built isolated systems without proper controls, leading to configuration drift across global hosts 20-22.
- **Action:** We can deploy the Red Hat Ansible Automation Platform integrated directly with Terraform workflows and GitLab to establish a single "golden path" for Configuration as Code (CaC) 7, 20, 22.
- **Value:** This establishes strict DevOps guardrails and ensures reliable, predictable deployments across all hybrid environments, optimizing cloud infrastructure costs and accelerating provisioning times for new digital services without vendor lock-in 7, 12, 20, 22.